

Applications on chipsets

Since the device uses system-on-chips, it is used not only for communication; applications built by Suvana Solutions can also be loaded on it. The application-design depends on a customer's requests and usage, and can be modified according to their needs. Different applications have been developed for domestic and industrial use, which are loaded on to the users' devices. An application enables you to use a control device and enter instructions like the threshold of power consumption, time settings for appliances like a geyser, etc. All the information gets stored on the application's network and the device is monitored using this information.

Smart Plugg can also disconnect the load automatically in some of the pre-defined cases, or by issuing on-demand cut-off command either from remote system or by SMS/e-mail. Varanasi elaborates, "The applications stay on the topmost layer and there are two applications running at anytime—a ZigBee application on the Smart Plugg device and a Linux/Windows/Symbian-based application on the control-level, i.e., on your PC, laptop, mobile or touch panel."

The average self-power consumption of Smart Plugg is in the range of 60mA for 100metre range model and higher for higher-range models. Since it measures energies of AC-connected loads, there is no necessity of operating it on battery.

The application areas

A Smart Plugg can be used with any electrical appliance at home—AC, refrigerator, geyser, etc.—which you feel should be monitored for energy consumption. "At present, the device costs approximately Rs 2500, but Suvana Solutions is working on a technology to bring the cost down to as low as Rs 500. This way, users will be able to use a Smart Plugg for any of the appliances that consumes power heavily. For instance, electricity bills in India are based on slabs. If you exceed



Inside of Smart Plugg

one slab, you can instruct your Smart Plugg to notify you and reduce your energy consumption on appliances like ACs and geysers. In some cases, this may also mean that an appliance like a refrigerator is faulty and you can get it checked much before your hefty bill arrives at your door," Varanasi shares.

In the hospitality industry Smart Pluggs can be attached to equipment like dishwashers and hot plates. Varanasi illustrates, "Say, Smart Pluggs are attached to five hot plates. The data stored on the network can be used to compare the difference in energy consumed by the various hot plates. If one of them is faulty, you'll know immediately and can get it repaired or replaced. This is where you make your return on investment on Smart Pluggs."

Suvana Solutions has also implemented the use of Smart Plugg in building management system in a skyscraper in Hyderabad. Varanasi comments, "Some people wash their cars daily and others don't wash it for months. Since the residents use a common tap at the parking basement, it is difficult to divide the cost of water used by an individual. It was unthinkable of giving a water connection from each home to the basement. So we came up with an idea of providing a Smart Plugg to all the residents

to monitor their water-usage in the basement. The tap could be used only after connecting the Plugg and each resident pays for the amount of water he consumes."

What lies ahead

Suvana Solutions is also developing a Smart Plugg solution to monitor modules in solar panels. "Solar plants are being deployed in huge areas and there is a lot of distance between each module. The cables are prone to damage due to rat bites but you do not get an alert immediately about a damage. The Smart Plugg will see a difference in the energy generated and quickly notify the network," Varanasi claims.

Besides, as mentioned earlier, Suvana Solutions also plans to introduce a low-cost Smart Plugg. The radio-frequency communication is being modified from comprehensive mesh to simple transceiver without losing key features. This will bring down the overall cost by eliminating the charge for proprietary license fee. They also plan to employ alternate low-cost communications for some applications and use fast-moving metering chips. Since the chip-level goes low, power is saved. ●

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